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Gegeven is de breuk $z = \frac{\sqrt{2}}{1+i}$. We zoeken de goniometrische vorm door eerst teller en noemer afzonderlijk om te zetten.

Teller $z_1 = \sqrt{2}$:

$$r_1 = \sqrt{(\sqrt{2})^2 + 0^2} = \sqrt{2}$$

$$\alpha_1 = 0 \Rightarrow z_1 = \sqrt{2} \operatorname{cis} 0$$

Noemer $z_2 = 1 + i$:

$$r_2 = \sqrt{1^2 + 1^2} = \sqrt{2}$$

$$\tan \alpha_2 = \frac{1}{1} = 1 \Rightarrow \alpha_2 = \frac{\pi}{4} \Rightarrow z_2 = \sqrt{2} \operatorname{cis} \frac{\pi}{4}$$

Quotient bepalen:

$$z = \frac{z_1}{z_2} = \frac{\sqrt{2} \operatorname{cis} 0}{\sqrt{2} \operatorname{cis} \frac{\pi}{4}}$$

$$z = \frac{\sqrt{2}}{\sqrt{2}} \operatorname{cis} \left(0 - \frac{\pi}{4} \right)$$

$$z = 1 \operatorname{cis} \left(-\frac{\pi}{4} \right)$$

$$\textbf{Antwoord: } z = \operatorname{cis} \left(-\frac{\pi}{4} \right)$$

References